

Technology Stack

Date	4 July 2026
Team ID	SWTID-2026-7907
Project Name	EduGenie – Google Gemini Powered Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2	Creates a responsive, interactive, and user-friendly interface for students.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, FastAPI	Provides high-performance APIs and integrates easily with Gemini AI services.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite (Development) / PostgreSQL (Production)	Stores user details, learning history, quizzes, summaries, and AI responses securely.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Google Cloud Platform	Scalable cloud hosting with reliable deployment and AI service integration.
5	Version Control & CI/CD	Git, GitHub	Enables version control, collaboration, code backup, and project management.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API, VS Code, Postman	Gemini API generates AI-powered educational responses. VS code is used for development , and Postman is used for API testing.